

Mimosa Reproduction

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1. Checking the code on the paper's data

Mimosa (Bi et al., Nucleic Acids Research 52:11455, 2024). Released model `training/model_mimosa.pth` on the paper's test sets `data/miRAW_Test0-9.txt` (548 positive + 548 negative pairs each), step size 5.

	Accuracy	PPV	Recall	Specificity	F1	NPV
Paper, Figure 3B (mean of 10 sets)	0.7568	0.6771	0.9332	0.5803	0.7932	0.8971
Released model, our run (mean of 10 sets)	0.7567	0.6897 (AP 0.6771)	0.9332	0.5803	0.7932	0.8972

The original code computes "PPV" with `average_precision_score` (AP) instead of precision. The paper's 0.6771 equals the AP implied by its own recall and specificity; the true precision is 0.690. We report both.

2. Changes to the original code

The 40-nt architecture, the training procedure and the test procedure are the authors'. Only `training/Mimosa.py` was modified:

Change	Why	Effect on the method
Command-line interface (<code>--mode</code> , <code>--data</code> , <code>--stepsize</code> , <code>--threads</code>), progress output, model loaded next to the script on CPU	Running the scripts on the cluster	None
PPV = <code>precision_score</code> ; AP printed as "paper PPV"	Original PPV was AP (Section 1)	Reported metric only
Authors' commented-out <code>torch.save</code> line enabled	Without it training saves no model; the paper selects the minimum-validation-loss model	None (restores the documented behaviour)
<code>--epochs</code> argument, default 40	Needed for the 120-epoch run	None at the default

Not changed: architecture (16 encoder blocks, 8 heads, dim 64), input encoding, base-pairing features (Smith-Waterman of miRNA nt 1-10 against the 30-nt core), loss (CrossEntropy on softmax output, as in the original), Adam lr $1e-4$ / weight decay $1e-5$, batch 256, dropout 0.1, seed 1234, test procedure. Training runs on CPU, as in the original code.

New files

Path	Purpose
<code>training/prepare_dataset.py</code>	Builds our dataset files (Section 3)
<code>training/make_site40_valsplits.py</code>	Training file: our 40-nt training sites + the 40-nt sites of the validation splits
<code>training/predict_site40.py</code>	Gene-level test on CPU by calling the authors' <code>get_cts</code> / <code>kmers_predict</code> / <code>decision_for_whole</code>
GPU scoring scripts in <code>training/</code> (listed in <code>WALKTHROUGH.md</code>)	The same gene-level test on GPU from precomputed window features (Section 5)
<code>training/metrics_by_kind.py</code>	Metrics by target type (Section 6)
<code>slurm/*.slurm</code> , <code>experiments/cpu_gpu_check/</code>	Slurm jobs; CPU/GPU agreement check
<code>WALKTHROUGH.md</code>	Every decision and command, in order

3. Data

3.1 Raw inputs (outside the repository)

Input	Content	Used for
<code>~/training_chunks/</code>	346 JSONL chunks, 34,560,002 rows, of which 11,987,208 miRNA-mRNA. A resampled training stream built from <code>data_with_negatives</code> <code>final_train</code> (each pair repeated 4.3× on average)	Training
<code>~/data_with_negatives/rna_rna/miRNA_mRNA/</code>	<code>final_valid_unseen_{pair,source,target}.jsonl</code>	Model selection
	<code>final_test_unseen_{pair,source,target}.jsonl</code>	Test (used once)
<code>~/fasta_files/rna.fa</code>	All RNA sequences (T→U)	Sequences

unseen_source = miRNAs never seen in training; **unseen_target** = targets never seen; **unseen_pair** = neither seen. Negatives are made by swapping the miRNA or the target of a positive pair, so the same miRNAs and targets appear with both labels. The train split of `data_with_negatives` is not used.

3.2 Training rows (`training/prepare_dataset.py`)

Step	Rows
miRNA-mRNA rows in <code>training_chunks</code>	11,987,208
Unique (miRNA, target, label)	2,761,271
Dropped: miRNA / target sequence not in <code>rna.fa</code>	78,345 / 51,556
Dropped: miRNA longer than 30 nt (Mimosa pads miRNAs to 30 nt)	30,147
Dropped: target with non-ACGU characters / shorter than 40 nt	1,751 / 333
Dropped: label conflicts / pairs also in a validation or test split	0 / 0
Kept	2,599,139
Used for training: pairs whose target is already a 40-nt site (Mimosa's training input)	39,770 (22,201 pos / 17,569 neg)

These sites have the same layout as the paper's training sites (30-nt core + 5-nt flanks, 5'→3'). Validation and test splits are kept in full, in their original order, without deduplication or resampling; a `mimosa_status` column marks rows Mimosa cannot process (sequence missing from `rna.fa`, miRNA outside 10–30 nt, non-ACGU).

Split	Rows	Processable	%	40-nt-site rows (processable)
valid_unseen_pair	12,200	11,281	92.5	180
valid_unseen_source	206,056	191,156	92.8	3,615
valid_unseen_target	189,364	179,660	94.9	2,579
test_unseen_pair	44,580	42,476	95.3	607
test_unseen_source	398,828	379,727	95.2	5,905
test_unseen_target	415,348	394,110	94.9	5,057

4. Training and model selection

Original `perform_train`, unchanged: 39,770 training sites (the paper uses 58,793 miRAW sites), validated each epoch on the 6,374 processable 40-nt-site rows of the three validation splits (3,419 pos / 2,955 neg). As in the paper, the model with the lowest validation loss is selected. 120 epochs (8 h 18 min on 24 CPU cores).

Epoch	Val loss	Acc	PPV	Recall	Spec	F1	NPV
36 (best of the first 40)	0.4826	0.823	0.840	0.828	0.818	0.834	0.804
118 (selected)	0.4784	0.830	0.849	0.831	0.829	0.840	0.809

Validation loss is flat after about epoch 36; the selected model differs little from a 40-epoch model.

5. Gene-level test

Procedure as in the paper: target reversed to 3'→5', cut into 40-nt windows with step 5 (`get_cts`), every window scored with the authors' features (`kmers_predict`), pair positive if any window scores above 0.5 (`decision_for_whole`). The test was run once, with the selected model.

For speed the windows were scored on GPU from precomputed features. On `valid_unseen_pair` (11,281 pairs) this gives the same decision as the original CPU code for every pair.

Split	Rows	n	Acc	PPV	AP	Recall	Spec	F1	NPV
test_unseen_pair	all, unprocessable = negative	44,580	0.517	0.510	0.509	0.866	0.168	0.642	0.556
	processable	42,476	0.513	0.510	0.509	0.900	0.118	0.651	0.539
test_unseen_source	all, unprocessable = negative	398,828	0.518	0.511	0.509	0.867	0.169	0.643	0.560
	processable	379,727	0.514	0.511	0.510	0.901	0.118	0.652	0.540
test_unseen_target	all, unprocessable = negative	415,348	0.514	0.508	0.507	0.864	0.165	0.640	0.548
	processable	394,110	0.511	0.508	0.508	0.903	0.112	0.651	0.534
Paper, miRAW	10 × 1,096		0.757	0.690	0.677	0.933	0.580	0.793	0.897

6. 40-nt sites versus transcripts

The same test predictions split by target type (decided before the test was run). A 40-nt target gives a single window, so its prediction is a site-level prediction.

Split	Target	n	Acc	PPV	Recall	Spec	F1	NPV
test_unseen_pair	40-nt site	625	0.818	0.855	0.782	0.857	0.817	0.784
	transcript	43,953	0.512	0.507	0.867	0.158	0.640	0.544
test_unseen_source	40-nt site	6,083	0.834	0.867	0.795	0.874	0.830	0.805
	transcript	392,707	0.513	0.508	0.868	0.158	0.641	0.546
test_unseen_target	40-nt site	5,593	0.864	0.840	0.903	0.826	0.870	0.893
	transcript	409,711	0.510	0.506	0.864	0.156	0.638	0.533

All rows, unprocessable = negative. 84 targets that are raw sequences of other lengths are left out of this table.

Negatives called positive by target length (valid_unseen_pair, selected model)

Target length (kb)	<0.5	0.5-1	1-1.5	1.5-2	2-2.5	2.5-3	3-3.5	3.5-4	≥4
Negatives	88	194	429	787	922	989	1,008	972	137
Called positive	0.20	0.78	0.88	0.91	0.93	0.90	0.93	0.92	0.91

On unseen pairs the model separates 40-nt sites well (accuracy 0.82–0.86, specificity 0.83–0.87). On transcripts it calls almost every pair positive: our targets have about 510 windows on average, and with that many windows at least one exceeds 0.5 for about 90 % of negatives.

7. Deviations from the paper

- Training data: 39,770 40-nt sites from our dataset instead of 58,793 miRAW sites.
- Validation: the 40-nt-site rows of our validation splits (the paper follows TargetNet's split). Epochs: 120 instead of 40; the selected epoch (118) is chosen by validation loss as in the paper.
- Test data: our unseen splits (several species, transcript targets up to 4,076 nt, swap negatives) instead of miRAW (human 3'UTRs). About 5 % of test rows cannot be processed and are reported both ways.
- Test windows were scored on GPU from precomputed features; decisions are identical to the original CPU code on the validation pairs checked.

8. Conclusions

- Mimosa trained on our data with the paper's method is at chance at gene level: accuracy 0.51–0.52 on all three test splits, with most negatives called positive.
- The model does learn to recognise binding sites: on 40-nt targets of unseen pairs, miRNAs and targets its accuracy is 0.82–0.86.
- The gap is consistent with applying a site classifier to whole transcripts with the any-window-above-0.5 rule: false positives grow with target length and reach about 90 % above 1 kb.